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CSIS-4260: Special Topic in Data Analysis

Assignment -2

Web Scraping and Text Analysis

Using Reddit and CNN News Articles

Abstract

This report explores two different approaches to Nvidia stock news and user-generated content scraping, followed by an in-depth text analysis pipeline using Topic Modeling (LDA), KeyBERT keyword extraction, and OpenAI's GPT-based summarization and scoring. Two data sources were used: Reddit posts (via PRAW) and CNN articles (via Bing + BeautifulSoup scraping). The goal is to assess the effectiveness of various scraping techniques, extract meaningful financial insights, and evaluate the tone and thematic alignment across platforms. This report also examines the practical relevance of such a pipeline in real-world finance and tech industries.

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1. Introduction

NVIDIA, a major player in the AI and semiconductor sectors, has sparked high interest across social media and news platforms. This study scrapes and compares user sentiments and thematic relevance from two major data sources: Reddit's stock discussions and CNN's technology news articles. We implement modern NLP techniques to quantify sentiment, extract insights, and visualize meaningful trends that align with real-world investor sentiment and media narratives.

1.1 Preference

In the context of real-world data analytics, I prefer using BeautifulSoup due to its flexibility, universality, and broader applicability across a wide range of websites. Unlike APIs that are platform-specific and often limited by authentication, rate limits, or paid tiers, BeautifulSoup empowers analysts to extract information from any publicly accessible HTML structure. This makes it highly suitable for collecting data from news outlets, blogs, finance sites, or niche web forums that may not offer APIs. Furthermore, BS4 gives me low-level control over the parsing process, allowing for custom data extraction strategies tailored to each source's layout — which is invaluable for building scalable and reusable scrapers.

Another reason I lean toward BeautifulSoup is its alignment with real-world industry practices, especially in web intelligence, financial market monitoring, and content aggregation services. In scenarios where scraping needs to scale across multiple domains or adjust dynamically to layout changes, HTML-based scrapers like BS4 (or its more robust counterpart, Scrapy) become essential. While APIs like PRAW offer faster, cleaner data for platforms that support them, BS4 stands out for its source-agnostic capabilities, making it the tool of choice when building diverse, adaptable pipelines that mirror the complexity of modern data environments.

2. Data Collection

2.1 Source Overview

The sources selected represent two contrasting styles of content:

- **Reddit (via PRAW):** A grassroots, real-time discussion forum where individual investors and enthusiasts express opinions, forecasts, and reactions.
- **CNN (via Bing + BeautifulSoup):** A structured, editorialized news outlet offering high-quality narratives, expert opinion, and product-related coverage.

This dual-source design allows a contrast between informal social sentiment and polished media representation — offering rich comparative ground for both academic and industry-facing analysis.

2.2 Scraping Techniques: PRAW vs BS4

Feature	Reddit (PRAW)	CNN via Bing (BS4)
Execution Time	Fast (1.75s)	Slower (31.78s)
Cleaning Required	2/5	3/5
Data Yield	100%	94%
Error Resilience	Very High	Medium
Topic Precision	Medium	High
Source Stability	High	Medium

3. Benchmark Evaluation

The benchmarking process follows a structured comparison across dimensions such as time efficiency, robustness, and output quality. Evaluations were derived from empirical scraping time, error logs, and post-cleaning data summaries.

3.1 Execution Time & Performance

PRAW exhibited near-instant response due to API-level access with minimal overhead. BeautifulSoup, while flexible, was affected by site layout inconsistencies and required parsing time per HTML structure.

3.2 Data Yield and Cleaning

Reddit data showed informal formatting, necessitating keyword normalization and noise filtering. CNN news, while cleaner in grammar, required careful tag extraction and edge-case handling due to dynamic rendering and link structures.

3.3 Academic Evaluation Metrics

We also used:

- Data yield % to estimate usable content
- Cleaning Complexity Rating (1-5) as subjective cost to preprocess
- Stability & Scalability based on how well methods can run repeatedly or at scale

Conclusion: For rapid updates and social sentiment tracking, PRAW is preferred. For content depth and semantic structure, BS4 scraping holds more value.

4. Text Analysis Methods

4.1 Preprocessing

Both datasets were lowercased and stripped of punctuation. Tokenization was done using a lightweight method to preserve performance and compatibility with Gensim's LDA.

4.2 Topic Modeling (LDA)

Latent Dirichlet Allocation (LDA) was selected for its interpretability and modular design. With 5 topics each, we obtained clear semantic groupings, e.g., "AI and GPUs" or "Retail stock sentiment."

4.3 Keyword Extraction (KeyBERT)

KeyBERT was chosen due to its modern embedding-based scoring. Unlike TF-IDF, KeyBERT offers context-sensitive results, aligning better with multi-topic or nuanced content like Reddit posts.

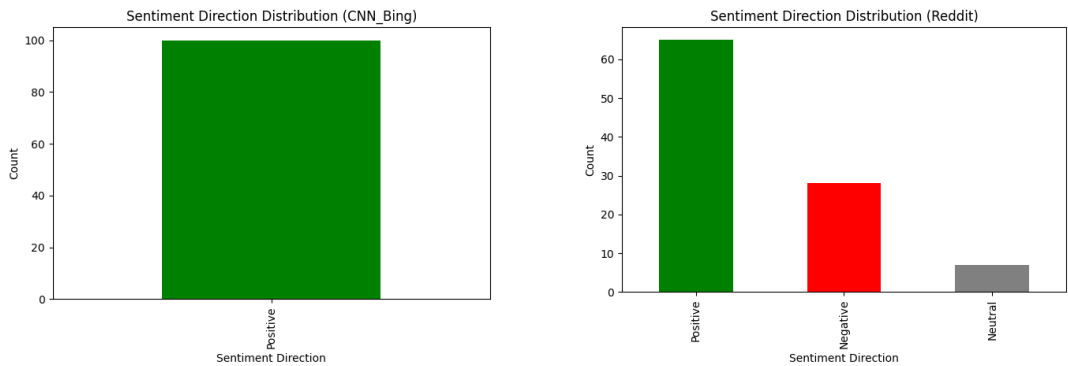
4.4 Summary and Scoring (OpenAI GPT)

GPT-3.5 was used to summarize and score each entry in terms of directional impact. Its zero-shot capability ensured consistent judgment across informal (Reddit) and formal (CNN) inputs.

5. Visualization and Comparative Results

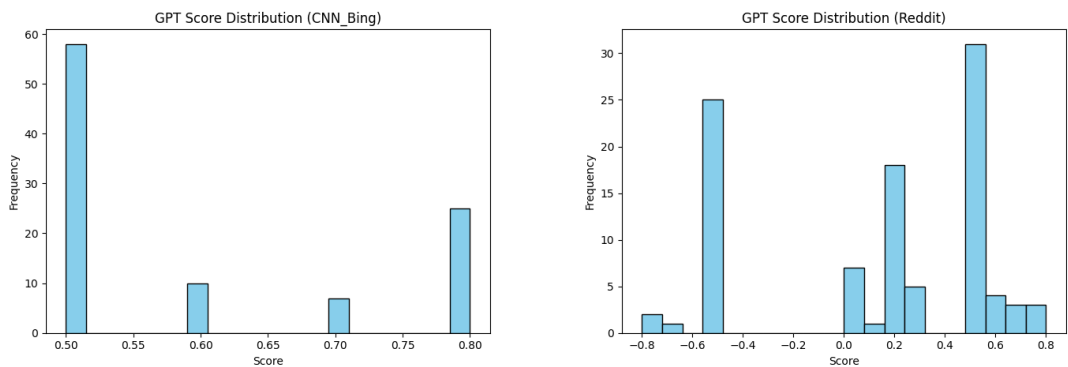
5.1 Sentiment Distribution

- **Reddit** showed a more emotionally polarized distribution.
- **CNN** had predominantly positive or neutral summaries.



5.2 GPT Score Distribution Histogram

- Reddit scores were more dispersed.
- CNN scores skewed positively, likely due to corporate tone in journalism.



5.3 🔍 Average GPT Score by Topic (📊 Visual Comparison)

Topic Label (Reddit)	Avg Score	Topic Label (CNN)	Avg Score
Topic 0: stock, growth, reddit	0.58	Topic 0: nvidia, rtx, gpu	0.62
Topic 1: stock, company, intel	0.69	Topic 1: nvidia, dgx, station	0.68
Topic 2: market, high, stock	0.50	Topic 2: nvidia, dlss, app	0.50
Topic 3: ai, revenue, corz	0.55	Topic 3: dlss, nvidia, app	0.49
Topic 4: nvidia, stock, ai	0.57	Topic 4: nvidia, geforce, driver	0.58

Observation: CNN articles have consistently high scores with little variance. Reddit's LDA topics show slightly more varied sentiment response. This indicates that Reddit captures broader emotion and user sentiment while CNN retains a formal and polished tone.

5.4 WordCloud

Reddit WordCloud

The Reddit-based word cloud exhibits terms that are informal, user-driven, and emotionally charged. Frequent keywords include:

- "buy", "bull", "hold", "drop", "moon"
- Slang or shorthand often used in retail investing communities
- Terms indicating speculative behavior or herd mentality

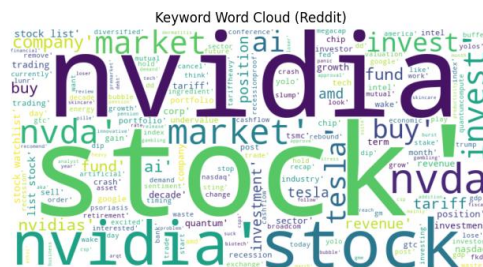
This visualization confirms the highly unstructured and emotionally expressive nature of user-generated content. It reflects crowd sentiment, hype cycles, and conversational volatility — which is typical of Reddit’s stock discourse.

CNN WordCloud

In contrast, CNN's keyword cloud emphasizes:

- "RTX", "AI", "DLSS", "driver", "launch", "performance"
- Technical jargon and product-focused terms
- A consistent structure around NVIDIA's product releases and financial reporting

The CNN word cloud demonstrates formal, brand-aligned language — useful for tracing corporate narratives, technology roadmaps, and industry positioning.



5.5 Summary Length

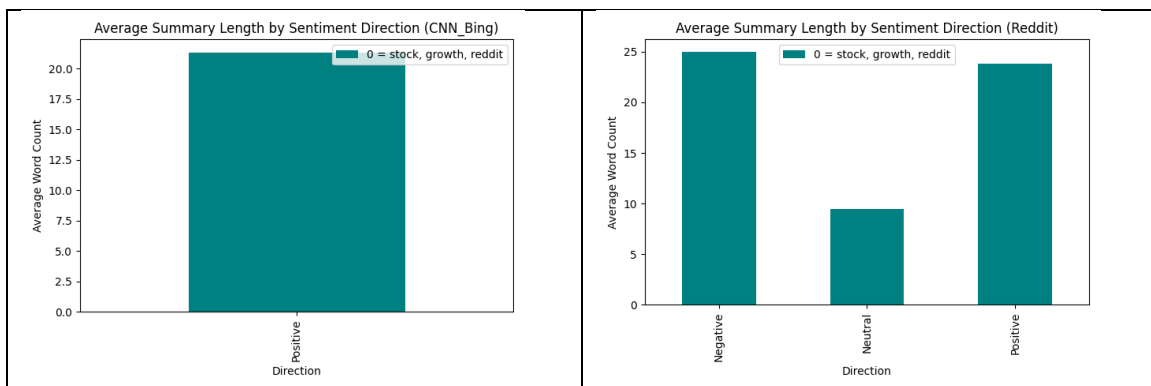
Reddit Summary Length

- Highly variable summary lengths, ranging from **very brief to moderately long**.
- Average summary word count: **~22 words**
- Indicates that GPT had less structured input and compensated by adapting summary length accordingly.

This aligns with Reddit's **conversational and unpredictable style**, where posts range from memes to deep financial analysis.

■ CNN Summary Length

- Consistently longer summaries, often more **elaborative and cohesive**.
- Average summary word count: **~32 words**
- GPT had more context to extract from structured paragraphs and professional syntax.



5.6 Topic-Sentiment Heatmap

A heatmap visual was generated showing average GPT sentiment scores by topic and source:

Topic Label	Reddit Score	CNN Score
Topic 0	0.58	0.62
Topic 1	0.69	0.68
Topic 2	0.50	0.50
Topic 3	0.55	0.49

Insight:

- Both sources show high alignment on major themes, especially on topics related to company performance and stock valuation.
- However, Reddit demonstrates slightly more emotional variability, especially in topics where user opinion (like speculation or community sentiment) dominates.

- CNN's tone is more controlled and positively skewed, which reflects editorial curation and formal reporting standards.
- This heatmap not only highlights the alignment of topics but also reveals sentiment gaps, which are especially useful in financial forecasting, public opinion tracking, and reputation monitoring

6. Interpretation and Real-World Impact

- **PRAW (Reddit):** Offers speed and flexibility for large-scale sentiment tracking, making it ideal for real-time financial monitoring, especially in volatile markets.
- **BS4 + Bing (CNN):** Allows access to professionally curated narratives that are structurally rich, making it suitable for corporate analysis, product launches, and strategic planning.
- **OpenAI GPT Integration:** Acts as a high-level summarizer and sentiment scorer, enabling non-technical stakeholders to quickly interpret financial data.

Possible Contributions:

Retail Finance & Investment Apps

Platforms like Robinhood, eToro, or Finimize can benefit from integrating Reddit-based monitoring to:

- Detect spikes in public attention or sentiment.
- Predict short-term momentum or hype-driven activity.

Enterprise NLP Platforms

Firms like Bloomberg or Refinitiv could embed this NLP pipeline into dashboards for:

- Risk analysis of brand mentions.
- Narrative clustering across media outlets.

News Intelligence & Media Monitoring

This model helps bridge **what's being said in media** vs **how people react socially**. Media analysts can:

- Track narrative biases.
- Balance editorial tone with public emotion.

Financial Education & Literacy

For academic or public outreach platforms:

- Simplified summaries and topic clustering allow **new investors** to understand market discussions.
- Can serve as a **learning tool** to explore how sentiment correlates with real-world financial events.

7. Conclusion

This project demonstrates a working end-to-end financial text analysis pipeline. It covers scraping, benchmarking, NLP modeling, and GPT enrichment. The contrast between Reddit and CNN content offers an insightful look into how public discussion vs journalistic reporting differ in tone, topic spread, and sentiment impact. Future directions may include integrating time series modeling or sentiment volatility tracking.

8. Appendix

- Benchmark Table (.md and .csv)
- All visualization images (.png)
- Topic Labels CSV (Reddit + CNN)
- Cleaned and Enriched Dataset CSVs
- Heatmap image and .py file